1. $F = (A \oplus B)(C+D)(A \odot C) + (A+B)(C'+D)(A+CC')$

$= (A'B+AB')(C+D)\overset{(AC+A'C')}{\cancel{(AB+A'B}} + (A+B)(C'+D)A'$

$= (A'B+AB')(AC+A'C')(C+D) + (AA'+BA')(C'+D)$

$= (A'BAC + A'BA'C' + AB'AC + AB'A'C')(C+D) +$

$(A'B)(C'+D)$

$= (AA'BC + A'A'BC' + AAB'C + AA'B'C')(C+D) +$

$A'BC' + A'BD$

$= (0 + A'BC' + AB'C + 0)(C+D) + A'BC' + A'BD$

$= A'BC'C + A'BC'D + AB'CC + AB'CD + A'BC' + A'BD$

$= 0 + A'BC'D + A'BC' + AB'C + AB'CD + A'BD$

$= A'BC'(D+1) + A'BC(1+D) + A'BD$

$= A'BC' + AB'C + A'BD$

$= ABC' + AB'C + A'BD$ (Ans)

CamScanner

2. $F = (x'+y+z')(x+y')(x'+z)$

$= (x'\cdot y\cdot z') + (x\cdot y') + (x'\cdot z)$

$F' = (x\cdot y'\cdot z) + (x'y) + (xz')$ (Ans)

3

SOP

$F(v,w,x,y,z) = wy + wx + x'y$

$= wy(v+v')(x+x')(z+z') + wx(v+v')$

$(y+y')(z+z') + x'y(v+v')(w+w')(z+z')$

$= wy(vx+vx'+v'x+v'x')(z+z') + wx(v+v')$

$(yz+yz'+y'z+y'z') + x'y(vw+vw'+v'w+v'w')$

$(z+z')$

$= wy(vxz+vxz'+vx'z+vx'z'+v'xz+v'xz'+$

$v'x'z+v'x'z') + wx(vyz+vyz'+vy'z+vy'z'+$

$v'yz+v'yz'+v'y'z+v'y'z') + x'y(vwz+vw'z+$

$v'wz+v'w'z+vwz'+vw'z'+v'wz'+v'w'z')$

$= wyvxz + wyvxz' + wyvx'z + wyvx'z' +$
$wyv'xz + wyv'xz' + wyv'x'z + wyv'x'z' + wxvyz +$
$wxvyz' + wxvy'z + wxvy'z' + wxv'yz +$
$wxv'yz' + wxv'y'z + wxv'y'z' + x'yvwz + x'yvw'z$
$+ x'yv'wz + x'yv'w'z + x'yvwz' + x'yvw'z' +$
$x'yv'wz' + x'yv'w'z'$

$= vwxyz + vwxyz' + vwx'yz + vwx'yz' + v'wxyz +$
$v'wxyz' + v'wx'yz + v'wx'yz' + vwxy'z + vwxy'z' +$
$v'wxy'z + v'wxy'z' + vw'x'yz + v'w'x'yz +$
$vw'x'yz' + v'w'x'yz'$

$= 11111, 11110, 11011, 11010, 01111, 01110, 01011,$
$01010, 11101, 11100, 01101, 01100, 10011, 00011,$
$10010, 00010$

$= \Sigma(31, 30, 27, 26, 15, 14, 11, 10, 29, 28, 13, 12, 19, 3, 18, 2)$
Ans

POS

: $F(v, w, x, y, z) = wy + wx + x'y$

$= (wy + wx + x')(wy + wx + y)$

$= (wy + x' + wx)(wy + y + wx)$

$= (wy + x' + w)(wy + x' + x)(y(w+1) + wx)$

$= (wy + w + x')(wy + 1)(y \cdot 1 + wx)$

$= (w(y+1) + x')(wy + 1)(y + wx)$

$= (w \cdot 1 + x')(y + wx)$

$= (w + x')(y + w)(y + x)$

$= (w + x' + vv')(y + w + w')(y + x + vv')$

$= (w + x' + v)(w + x' + v')(y + w + v)(y + w + v')$
$(y + x + v)(y + x + v')$

$= (w + x' + v + yy')(w + x' + v + y')(y + w + v + xx')$
$(y + w + v' + xx')(y + x + v + ww')(y + x + v' + ww')$

$= (w + x' + v + y)(w + x' + v + y')(w + x' + v' + y)(w + x' + v' + y')$
$(y + w + v + x)(y + w + v + x')(y + w + v' + x)(y + w + v' + x')$
$(y + x + v + w)(y + x + v + w')(y + x + v' + w)(y + x + v' + w')$

CamScanner

$$= (w+x'+v+y+zz')(w+x'+v+y'+zz')(w+x'+v'+y+zz'$$
$$(w+x'+v'+y'+zz')(y+w+v+x+zz')(y+w+v+x'+zz'$$
$$(y+w+v'+x+zz')(y+w+v'+x'+zz')(y+x+v+w+zz'$$
$$(y+x+v+w'+zz')(y+x+v'+w+zz')(y+x+v'+w'+zz'$$

$$= (w+x'+v+y+z)(w+x'+v+y+z')(w+x'+v+y'+z)$$
$$(w+x'+v+y'+z')(w+x'+v'+y+z)(w+x'+v'+y+z')$$
$$(w+x'+v'+y'+z)(w+x'+v'+y'+z')(y+w+v+x+z)$$
$$(y+w+v+x+z')(y+w+v+x'+z)(y+w+v+x'+z')$$
$$(y+w+v'+x+z)(y+w+v'+x+z')(y+w+v'+x'+z)$$
$$(y+w+v'+x'+z')(y+x+v+w+z)(y+x+v+w+z')$$
$$(y+x+v+w'+z)(y+x+v+w'+z')(y+x+v'+w+z)$$
$$(y+x+v'+w+z')(y+x+v'+w'+z)(y+x+v'+w'+z')$$

$$= (v+w+x'+y+z)(v+w+x'+y+z')(v+w+x'+y'+z)$$
$$(v+w+x'+y'+z')(v'+w+x'+y+z)(v'+w+x'+y+z')$$
$$(v+w+x'+y'+z)(v'+w+x'+y'+z')(v+w+x+y+z)$$
$$(v+w+x+y+z')(v'+w+x+y+z)(v'+w+x+y+z')$$
$$(v+w'+x+y+z)(v+w'+x+y+z')(v'+w'+x+y+z)$$
$$(v'+w'+x+y+z')$$

CS CamScanner

= 00100, 00101, 00110, 00111, 10100, 10101,

10110, 10111, 00000, 00001, 10000, 10001,

01000, 01001, 11000, 11001.

$= \pi(4, 5, 6, 7, 20, 21, 22, 23, 0, 1, 16, 17, 8, 9,$

$24, 25)$.

~~Ans~~

CS CamScanner